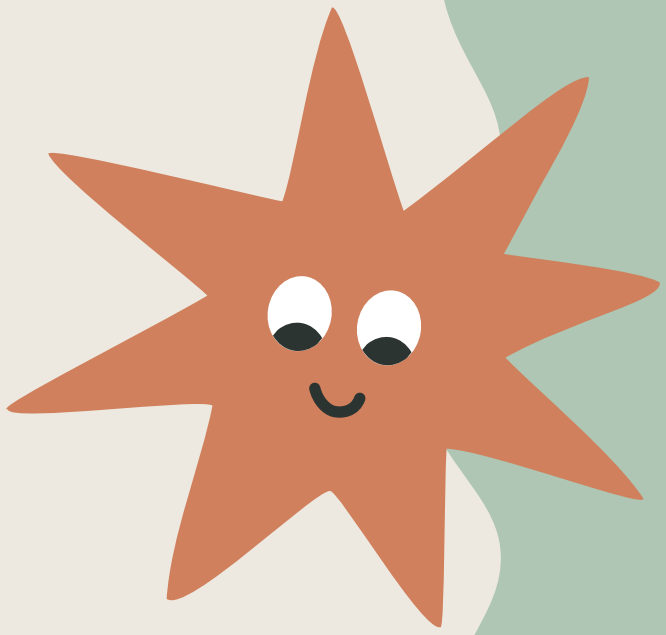
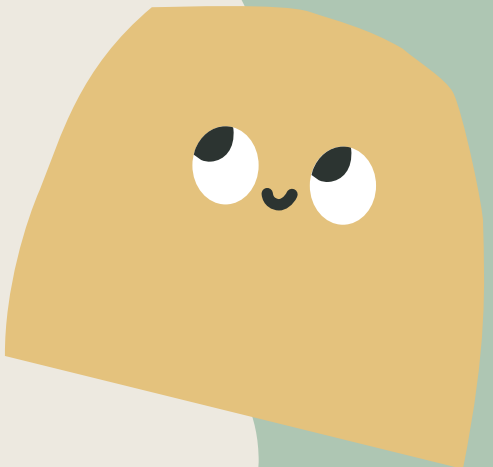
Knowledge Representation in AI

Lecture 5





Introduction

- How does an AI system represent what it knows about the world?
- Consider these statements:
  - Ali is a student
  - Ali studies Computer Science
  - All Computer Science students study programming
- A human can easily understand these statements and connect them
- But a computer cannot simply understand human knowledge in the same way we do
- We need a structured way to represent this information
- This is called **Knowledge Representation**

Prepared By: Ehtisham Rasheed

What is Knowledge?

- Knowledge is information that an intelligent system can use to understand a situation, reason about it, and make decisions
- For example:
  - Lahore is in Pakistan
  - This is information
- But if an AI system knows:
  - Lahore is in Pakistan
  - Pakistan is in Asia
- It can drive:
  - Lahore is in Asia

Prepared By: Ehtisham Rasheed

Data, Information, and Knowledge

- Data
  - Raw facts or values
  - For example: 25
  - By itself, it tells us very little
- Information
  - Data with meaning
  - For example: Ali's age = 25
  - Now the value has meaning
- Information together with relationships, rules, or understanding that can support reasoning
- Data → Information → Knowledge → Reasoning → Decision

Prepared By: Ehtisham Rasheed

What Is Knowledge Representation?

- **Knowledge Representation (KR)** is the process of representing information about the world in a form that an AI system can use for:
  - Reasoning
  - Decision making
  - Problem solving
  - Answering questions
  - Drawing conclusions
- Knowledge Representation = Giving knowledge to an AI system in a structured form so it can reason about it

Prepared By: Ehtisham Rasheed

Example

- Human knowledge
  - It is raining
- AI representation:
  - Rain
- Another statement
  - If it is raining, the road is wet
- Can be represented as:
  - Rain → WetRoad
- The AI can then use these statements to reason

Prepared By: Ehtisham Rasheed

Knowledge Base (KB)

- A **Knowledge Base** is a collection of knowledge represented in a form that an AI system can use
- It may contain:
  - Facts
  - Rules
  - Relationships
  - Constraints
  - Other domain knowledge
- The AI can use this knowledge when answering questions or making decisions

Prepared By: Ehtisham Rasheed

Facts

- A **fact** represents something that the system knows to be true
- Example:
  - Ali is a student
  - Lahore is in Pakistan
  - The temperature is 30°C
- Facts provide the basic knowledge from which reasoning can begin

Prepared By: Ehtisham Rasheed

Rules

- A **rule** describes a relationship or condition that allows the system to derive something
- Example:
  - If a person is a student, then the person is a learner
  - If the temperature is below zero, water may freeze
- Conceptually
  - Student → Learner
  - Low Temperature → Possible Freezing
- Rules allow an AI system to move from known information toward conclusions

Prepared By: Ehtisham Rasheed

What Is Inference?

- **Inference** is the process of deriving new information from existing knowledge
- Inference means reaching a conclusion using what is already known
- Known:
  - Ali is a student
  - All students are learners
- Inference
  - Ali is a learner
- The system did not necessarily receive the final statement directly
- It derived it from existing knowledge

Prepared By: Ehtisham Rasheed

Different Ways to Represent Knowledge

- There is no single way to represent knowledge
- AI can use different knowledge-representation techniques including:
  1. Logic
  2. Semantic networks
  3. Frames
  4. Rules
  5. Ontologies
  6. Knowledge graphs

Prepared By: Ehtisham Rasheed

Different Ways to Represent Knowledge

Technique	Main Idea
Logic	Formal statements and reasoning
Rules	IF–THEN relationships
Semantic Network	Concepts connected by relationships
Frames	Structured description of objects/concepts
Ontology	Formal domain concepts and relationships
Knowledge Graph	Entities connected through relationships

Example

- A university knows that Ali is a student. Every student must study at least one course. Ali studies Artificial Intelligence
- What are facts in paragraph above?
  - Ali is a student
  - Ali studies AI
- What is the rule?
  - Every student studies a course

Prepared By: Ehtisham Rasheed